

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Group ID: 25-26J-090

Project Title:

REAL-TIME AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT MANAGEMENT IN STREAMING ENVIRONMENTS

1. **IT22549136:**
AN AGENTIC MULTI-MODAL UNSTRUCTURED DATA PREPROCESSING FRAMEWORK WITH LLM-GUIDED ORCHESTRATION AND QUALITY-DRIVEN VALIDATION
2. **IT22893666:**
A SCALABLE FRAMEWORK FOR STRUCTURED DATA CLEANING IN STREAMING ENVIRONMENTS: AN ML-BASED PREDICTION APPROACH WITH HUMAN-IN-THE-LOOP GOVERNANCE
3. **IT22556684:**
AUTONOMOUS FRAMEWORK FOR REAL-TIME PII DETECTION AND PRIVACY PRESERVATION IN STREAMING DATA PIPELINES
4. **IT22204752:**
A FRAMEWORK FOR SCHEMA DRIFT DETECTION AND GOVERNED ADAPTATION IN STREAMING DATA PIPELINES

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		IT22204752	IT22556684	IT22893666	IT22549136
Data Pipeline	Data Sources:	A scenario-based corpus of 160 schema-drift instances across five domains: invoice (47), user profile (33), sensor (32), retail (30), and geo-sales (18). Datasets sourced from Kaggle and modified. 2,512 candidate column pairs generated (819 positive renames, 1,693 negative). Inputs accepted as JSON, CSV, or Kafka streams.	Publicly available PII datasets from Kaggle and Hugging Face, augmented with synthetic PII generated under realistic streaming conditions using Apache Airflow and Kafka schedulers. Accepts structured (DBs, CSVs), unstructured (PDFs, docs, text), and real-time streaming (Kafka, IoT) inputs.	Nearly 20 Kaggle datasets ingested via Apache Kafka in configurable batches of 500 records/batch. Covers healthcare, finance, IoT, and e-commerce domains. Synthetic datasets with known ground-truth quality annotations are used to evaluate correction accuracy.	Web-sourced text corpus (HTML-tagged, multilingual, clean, and noisy variants) and document-scan images of natural photographs. Datasets sourced from Kaggle and modified. Inputs accepted as plain text strings, delimited files (CSV, JSONL, TXT), or image files via the /ingest REST endpoint.
	Data Preprocessing:	Column name normalization applied before comparison: lowercase conversion, replacement of non-alphanumeric characters with underscores, collapsing of consecutive underscores, and stripping of leading/trailing	Dual detection strategy: (1) XLM-RoBERTa fine-tuned for multilingual PII detection using BIO tagging with confidence threshold 0.85, supporting PERSON, LOCATION, ORG, EMAIL, MARITAL_STATUS, PASSWORD, SEXUAL_ORIENTATION; (2) Deterministic regex	LLaMA 3.3-70B (via Groq API) automatically extracts validation rules from unstructured domain documents (PDFs, plain-text) into machine-readable CSV format, capturing data types, numerical ranges, categorical constraints, and compound conditions. Multi-dimensional issue detection: missing values (null/undefined/empty),	For text: LLM (locally-hosted Ollama via LangChain) selects and sequences 8 agents per input: html_strip, lang_detect, translator, spell, sentence_split, tokenizer, ner, validation. Canonical ordering is enforced regardless of LLM output. Fallback plan [spell, tokenizer, lang_detect] activates on LLM failure (2-5% of cases). For images: a rule-based decision engine computes Laplacian

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		underscores. Observed schema inferred directly from incoming batch data using conservative type inference (Boolean > Integer > Float > Numeric string > Datetime string > String). Structural diff identifies additions, removals, type changes, and nullability changes.	engine for structured PII (NIC, credit cards via Luhn algorithm, Sri Lankan bank prefixes, phone normalization). Priority-based entity merging: regex assigned priority 100 overrides NER for overlapping spans.	outliers (IQR analysis + rule-based), and invalid categorical values. Inter-column dependency analysis using chi-square tests (categorical), mutual information (mixed), and Spearman correlation (ordinal/numerical).	variance (blur < 120), mean luminance (brightness outside [70, 200]), and standard deviation (contrast < 60) to select agents.
	Data storage and versioning	The schema registry stores versioned schema snapshots identified by content hash; every schema state is uniquely addressable. Drifted batches are isolated in a staging queue pending human review; drift-free data proceeds directly to production. Audit log records every material event (append-only): drift summary, rename	No raw PII is committed to any storage layer. Only masked, encrypted, hashed, or tokenized records are written to protected storage. ChromaDB persistent vector database stores regulation document embeddings for RAG-based policy enhancement. Full audit trail logged per transformation with method identifier, timestamp, and	Persistent batch storage with atomic file writes ensures fault-tolerant recovery without data loss. Session-based state maintains the active rule set, dependency graph, trained model instances, and issue log across processing stages. Structured audit log with per-record granularity captures: original value, suggested value, final applied value, decision type (approve/reject/override/auto), responsible actor, model	Processed text and image records persisted to MongoDB using record UUID as document <code>_id</code> ; upsert semantics prevent duplicates on reprocessing. Collection indexed on {type, timestamp}. Final processed images saved to disk under <code>runs/{request_id}/final/output.png</code> ; intermediate candidates optionally saved under <code>runs/{request_id}/candidates/</code> for audit. SSE buffer (FIFO, 200 entries) streams results to React frontend in real time.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		mappings with confidence scores, risk score and contributing factors, routing decision, reviewer identity, timestamp, and stated reason.	regulatory citation (GDPR, HIPAA, CCPA, PCI-DSS v4.0, PDPA No. 9/2022).	confidence score, and timestamp.	
Model Development	Model Selection	Eight-feature probabilistic rename classifier combining four evidence types: lexical (edit distance, token overlap, Jaccard similarity), semantic (sentence-embedding cosine similarity to capture synonym substitution and concept-level renames), structural (type compatibility), and statistical (null-rate difference, mean difference, std difference, range overlap from baseline profile). Dual-mode risk scoring: Option A (dataset-level weighted formula: $\text{risk_A} =$	Fine-tuned XLM-RoBERTa model for multilingual PII NER with BIO tagging. Composite risk scoring: $\text{RS} = 0.7 \times \text{S} + 0.3 \times \text{C}$ (S: sensitivity weight LOW=1, MEDIUM=3, HIGH/CRITICAL=5; C: context weight logging=1, storage=2, analytics=3, external_transfer=5). Protection method selected by RS threshold: Masking ($\text{RS} \leq 2.0$), SHA-256 Hashing (2.0-3.5), AES-256-GCM Encryption (3.5-4.5), Tokenization ($\text{RS} > 4.5$), BCRYPT (passwords always), Generalization (low-risk DATE/LOC). RAG	Random Forest Classifiers for categorical attributes and Random Forest Regressors for continuous numerical attributes, selected for interpretability, overfitting resistance (bootstrap aggregation), and native feature importance estimation. SHAP and LIME provide prediction-level explainability. Composite quality score: completeness (40%, missing value ratio), validity (40%, outlier and invalid entry ratio), consistency (20%, duplicates and low-variance attributes). Score normalized 0-95 and mapped to grade scale A-F.	Locally hosted Ollama LLM (via LangChain Ollama interface) for text agent planning: inspects input and produces JSON array of agent names; acts strictly as a classifier, never generates text. spaCy en_core_web_sm for NER (PERSON, ORG, GPE, PRODUCT, DATE, TIME, LAW, EVENT). langid.py (Naive Bayes, 97 languages) for language detection. Rule-based decision engine for images: ROI agent (Canny edge + contour crop), Deblur (Wiener deconvolution, PSF as uniform box kernel), Enhancement (gamma + CLAHE on LAB L-channel, tile 8x8, clip 2.0), Denoise (NLMeans default). Post-transformation validation: SSIM ≥ 0.995 and PSNR ≥ 15.0 dB; revert if either fails.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		clip(0.55T + 0.20C + 0.20S + 0.05K + 0.10U, 0, 1)) and Option B (field-level union-of-risks: risk_B = 1 - product(1 - r_i)). Severity scale: deletion 1.0, type change 0.9, rename 0.7, nullability 0.5, addition 0.4.	subsystem (ChromaDB + Google Gemini) dynamically generates policy rules from uploaded regulation PDFs.		
	Model Training	Rename model trained on 160 schema-drift scenarios with scenario-level train-test split (120 train / 40 test), yielding 1,824 training pairs and 633 test pairs (819 positive, 1,693 negative). Operating threshold 0.5359 determined via precision-recall analysis. Results: precision 0.80, recall 0.90, F1 0.85, PR-AUC 0.9226, accuracy 0.8957. Heuristic weighted fallback is available when no trained model is	XLM-RoBERTa fine-tuned for multilingual PII detection on Kaggle and Hugging Face PII benchmark datasets augmented with synthetic streaming data. PyTorch with CUDA support. Weighted F1 = 0.913 across all entity types; 100% precision for NIC (full format + day-of-year validation) and credit card (Luhn algorithm). Hybrid ensemble (NER + regex) recovered ~11% missed entities, improving overall recall by 0.08 over NER-only. RAG-	Random Forest models trained on dependency-informed feature subsets per attribute using Scikit-learn. Preprocessing: median imputation for missing predictors, one-hot encoding for nominal categorical features. Dependency graph drives feature selection; only statistically significant dependencies (above a configurable threshold) are retained to prevent overfitting. Evaluated against three baselines: rule-only, statistical imputation (mean/mode), and batch processing equivalent. Quality score calibration validated via Pearson	No explicit model retraining performed. Ollama LLM used as a pre-trained planner without fine-tuning; agent selection relies on in-context prompting. spaCy en_core_web_sm and langid.py are deployed as pre-trained models. Image quality thresholds (blur < 120, brightness [70, 200], contrast < 60) calibrated from representative document scans and natural photographs; all thresholds exposed as configurable parameters. Validation thresholds (SSIM >= 0.995, PSNR >= 15.0 dB) established from per-agent quality metric distributions.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		present. Severity thresholds and risk scoring weights are configurable parameters.	generated policy rules matched expert rules in 89% of cases by semantic similarity.	correlation between composite score and actual proportion of clean records.	
Model Integration	Tools used	Python; Apache Kafka and Zookeeper (JSON/CSV/Kafka input routing); schema registry for versioned baseline management; probabilistic classifier (scikit-learn or equivalent) for rename model; sentence embedding model (e.g., all-MiniLM-L6-v2 via SentenceTransformers) for semantic feature extraction; structured audit logging.	FastAPI (Python async backend); React 19 + Vite (frontend with entity visualization, pie chart, manual override, profile loading); PyTorch + HuggingFace Transformers (XLM-RoBERTa NER); ChromaDB (vector DB for regulation documents); Python Cryptography library (AES-256-GCM); Google Gemini API (RAG policy generation); SpaCy (sentence segmentation); SentenceTransformer all-MiniLM-L6-v2; PyPDF2 (text extraction); Apache Kafka (streaming ingestion); Apache Airflow (synthetic data generation).	Apache Kafka + Zookeeper (Confluent Platform); FastAPI (REST API orchestration, session state); React + Vite (frontend: data upload, rule review, quality report visualization); Scikit-learn (Random Forest, preprocessing pipelines); Groq API with LLaMA 3.3-70B (rule extraction); SHAP and LIME (explainability); pypdf and tabula-py (PDF/document parsing); Pandas and NumPy (data manipulation).	FastAPI (REST and SSE API, /ingest endpoint); Apache Kafka (5 topics: raw_input, detected_text, detected_image, processed_text, processed_image); MongoDB (persistence, UUID as _id); React (SSE frontend via EventSource, /results polling); Ollama + LangChain (local LLM hosting and orchestration); spaCy en_core_web_sm (NER); NLTK sent_tokenize and word_tokenize; langid.py (language detection); pypellchecker; OpenCV (image processing, Laplacian, Canny, CLAHE); NumPy (fallback image computations); Google Cloud Translation API; Pydantic (schema validation).

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Model Deployment	Testing Environments	Held-out evaluation on 633 test pairs (40 unseen scenarios) using scenario-level split to prevent leakage. Evaluated against three progressively stronger baselines: name similarity only (F1 0.67), lexical+type (F1 0.75), lexical+semantic (F1 0.82), full model (F1 0.85, PR-AUC 0.92). Risk scoring validated across five severity categories. Governance workflow tested with approve, reject, and rollback actions across schema version history.	Evaluated on hybrid dataset: publicly available PII benchmarks (Kaggle, Hugging Face) + synthetic PII under streaming conditions. Tested against five existing tools (Protecto AI, Apache Airflow, Dagster, DataGrail, OneTrust) - the proposed framework is the only solution eliminating raw PII before persistence. Leakage score = 0.00 for AES-256-GCM, tokenization, and bcrypt. Latency: 38 ms/document for AES-256-GCM, 12 ms for masking (documents up to 50 KB). Throughput: 1,200 documents/minute on 8-core machine.	Assessed across five dimensions: (1) rule extraction precision/recall on manually annotated corpora across three domains; (2) issue detection F1-score per category (missing, outliers, invalid); (3) correction accuracy: MAE for numerical, classification accuracy for categorical, evaluated on synthetically corrupted datasets with known ground truth; (4) throughput and latency under varying input rates/batch sizes via Kafka; (5) quality score calibration via Pearson correlation. Compared against rule-only, statistical imputation, and batch processing baselines.	Text: 88-92% cleaning success rate; mean overall confidence 0.84 (std 0.11); LLM plan failure 2-5% (fallback activated). Agent utilization: lang_detect + tokenizer >95%, html_strip 35-45%, spell invoked 40-60% (15-20% skipped via noise ratio), ner 70-85% of English records, translator dynamically inserted 20-30%. Latency reduction: 50% (English HTML + spell errors, 4 agents), 62% (clean German, 3 agents), 87% (clean English, 1 agent), mean 35-40% vs. full pipeline. Image: 92% transformation acceptance; SSIM range 0.994-0.998; 8% reverted (mostly over-enhancement). End-to-end latency: text 0.6-2.5s, image 2.2-5.4s. Throughput: text 6-8 rec/s, image 0.8-1.5 rec/s.
	Deployment Platform	Plug-in service integrated into existing batch or streaming infrastructure without requiring pipeline	FastAPI async backend for API orchestration; Apache Kafka for real-time streaming ingestion; React 19 + Vite	Microservices-inspired layered architecture: React+Vite frontend tier, FastAPI backend exposing RESTful endpoints, Confluent Platform	FastAPI REST/SSE API layer; Apache Kafka streaming layer (5-topic pipeline with independent consumer threads for text and image); MongoDB persistence

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		modifications. Accepts data streams via JSON, CSV, or Kafka topics. The schema registry and audit log operate as persistent components. Deployable as a standalone microservice or embedded within existing data governance tooling.	frontend (entity detection dashboard, policy rules page, protected document download); ChromaDB as persistent vector store for regulation documents. Supports cloud and on-premise deployments; GPU-accelerated NER via PyTorch CUDA where available.	Kafka+Zookeeper for enterprise-grade message durability and consumer group management, Scikit-learn ML engine, Groq API LLM module. Supports horizontal scaling through increased Kafka partitions and consumer instances.	layer; React frontend consuming SSE stream at /events and polling /results on load. Text throughput bottleneck: LLM inference (Ollama). Image throughput bottleneck: NLMeans denoising (800-1800ms) and Wiener deconvolution (600-1400ms). Kafka routing latency: 20-100ms per message.
	Deployment Method	Four-stage sequential pipeline: (1) schema drift detection via column normalization + structural diff; (2) rename resolution via 8-feature probabilistic classifier; (3) dual-mode risk scoring (Option A dataset-level or Option B field-level); (4) governance-based routing with unconditional staging for any detected drift. Human reviewer takes one of three actions:	Five-layer inline pipeline at ingestion point: Data Ingestion and Buffering → Multi-Method PII Detection (NER + regex in parallel) → Entity Merging and Deduplication → Policy Engine and Risk Assessment → Adaptive Protection Module → Downstream Delivery. Protection applied in longest-first entity ordering to prevent span conflicts. RAG-based policy enhancement	Seven-stage modular pipeline: (1) Data ingestion via Kafka consumer or direct file upload; (2) LLM rule extraction and preprocessing; (3) Inter-column dependency analysis; (4) Multi-dimensional issue detection; (5) Dependency-informed ML correction; (6) HITL validation (confidence-based escalation: $\geq 85\%$ auto-apply, $< 85\%$ queued; 30-second timeout per correction; approve/reject/manual override); (7) Composite quality scoring and report generation.	Kafka router-consumer pattern: /ingest endpoint enqueues to raw_input topic; router worker inspects payload type and forwards to detected_text or detected_image topic; separate consumer threads invoke AgenticTextProcessor (LLM planning + agent execution) or ImagePreprocOrchestrator (rule-based decision + validate-then-commit per agent); results worker normalizes records, upserts to MongoDB, and appends to SSE buffer. Text files split row-by-row at ingestion for parallel Kafka messaging. Consumer groups

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		approve and promote, reject, or rollback to a previous schema version by content hash.	pipeline: PyPDF2 extraction → SpaCy semantic chunking (500-char windows, 100-char overlap) → SentenceTransformer embedding → ChromaDB storage → Gemini policy generation.	Session-based state is maintained throughout.	enable horizontal scaling without processing logic changes.
Future Enhancements	Model Improvement:	Larger evaluation corpus (hundreds of columns, broader domain naming conventions) for production-ready rename model. Retrieval-and-reranking upgrade following ReMatch architecture to reduce errors in the ambiguous mid-score region. Multi-step governance workflow for regulated datasets: technical impact assessment followed by compliance confirmation. Audit log infrastructure already	Apache Spark integration for enterprise-scale orchestration. Homomorphic encryption to enable compute-on-encrypted-data analytics. Adaptive differential privacy budgets. Federated variant for edge-node processing without centralizing raw data. Extension to additional regulatory frameworks beyond current GDPR, HIPAA, CCPA, PCI-DSS, and PDPA coverage.	RAG-based rule extraction to ground derivation in domain-specific corpora and reduce LLM hallucination risk. Online learning for correction models to adapt incrementally to concept drift without full retraining. Joint consistency enforcement (constraint satisfaction layer) to address multi-constraint consistency gap relative to HoloClean. Differential privacy integration for regulated domain deployment. Multi-broker Kafka evaluation for enterprise-scale horizontal scalability.	Learned deblurring model to handle motion blur (directional PSF), replacing current uniform box-kernel Wiener deconvolution. Timeout-based LLM fallback with graceful degradation for slow-but-successful Ollama responses. Feedback loop tracking which agent plans produce highest downstream model performance on held-out validation sets. Extension to audio preprocessing (SNR and clipping-rate driven agent selection) and tabular data preprocessing (column-level statistical profile driven imputation strategy).

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

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